

Rashmi Kant

Ph.D. in Computer Science and System Engineering | Mechanical Engineering

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Professional Summary

- Interdisciplinary researcher with expertise in **bio-inspired robotics, computational mechanics, Multiphysics coupled modeling, and multidisciplinary design optimization**.
- Experienced in developing custom computational frameworks for the design and analysis of **micro/nano aerial robots and composite aircraft**, integrating **Multiphysics coupled analysis**.
- Developed **Physics-AI and surrogate modeling approaches** for engineering prediction and optimization, with strong programming expertise in Python, Fortran, and C/C++.
- Experienced with advanced FEM/CFD simulation tools as well as microfabrication and experimental development of bio-inspired robotic systems.
- Research interests center on combining **physics-based modeling, AI, and optimization to develop innovative and sustainable next-generation robotic and aerospace systems**.

Core Expertise

Bio-inspired micro-flying robotics	Plant Inspired micro-flying robotics	multidisciplinary design optimization (MDO)	Multiphysics coupled analysis
composite aerostructures	GPU acceleration	AI / ML / Physics-AI	Digital Twins

Professional Experience

1. **Senior Research Scientist** / *Morgenrot.net*, Japan | Jun 2025 – Present
 - Optimizing CAE/CFD simulations (OpenFOAM, ANSYS) and ML/DL workflows (PyTorch, TensorFlow) on GPUaaS cloud for high performance.
 - Enhancing GPU job scheduling, automation, and resource utilization while driving research in HPC, AI-driven simulations, and business expansion.
2. **Researcher (contract)** / *Mque.co.jp*, Japan | April 2025 – May 2025
 - Performed supersonic/hypersonic CFD analyses and thermal–chemical non-equilibrium simulations using SU2/SU2-NEMO.
 - Developed a fundamental **optimization algorithm** of aerodynamic structure in a **supersonic/hypersonic** flow field.
3. **Researcher** / *Institute of Fluid Science, Tohoku University*, Japan | Jan 2023 – March 2025
 - **Developed custom codes for optimization of aero-structural coupling that include buckling tolerance analysis** and design optimal structure of **composite aircraft**.
 - **Design and develop a micro air vehicle using two-way coupled analysis**.
4. **Research Associate** / *Kyushu Institute of Technology*, Japan | Oct 2022 – Jan 2023
 - Innovated bio-inspired Nano air vehicle designs using **Multiphysics-coupled analysis** (fluid-structure-piezoelectric-circuit) and pioneered **Computational Biomimetics**.

- Innovated design solutions for operational challenges in constrained environments.
 - 5. **Doctoral Research** / *Kyushu Institute of Technology, Japan* | Oct 2022 – Jan 2023
 - Development of Polymer Micromachined Flapping-Wing Nano Air Vehicle (FWNAV)
 - Design flyable FWNAV using **iterative design window search methodology**, linear and **nonlinear transient structural analysis**, and **fluid-structure interaction analysis**.
 - Developed using innovative fabrication techniques, polymer micromachining.
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Education

- **Doctor of Philosophy (Ph.D.) in Mechanical Information Science and Technology**
Kyushu Institute of Technology, Japan | 2019-2022 | GPA: 3.9/4.0
 - **Master of Technology (M.Tech) in Materials Science and Technology**
National Institute of Technology, Durgapur, India | 2015-2017 | CGPA: 8.41/10
 - **Bachelor of Technology (B.Tech) in Mechanical Engineering**
Institute of Technology, Bilaspur, India | 2010-2014 | CGPA: 8.64
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Technical Skills

- **Programming:** Python, MATLAB, Fortran, C/C++
 - **AI / ML:** PyTorch, TensorFlow, Physics-AI, data-driven modeling, machine learning
 - **CFD / Thermal-Fluid:** OpenFOAM, ANSYS, SU2, SU2-NEMO, OpenLB, DualSPHysics
 - **Structural / Multiphysics:** NASTRAN, ABAQUS, PATRAN, CATIA V5, Star CCM
 - **Meshing / Visualization:** Pointwise, GMSH, ParaView, VTK
 - **Optimization:** Multiobjective optimization, surrogate-assisted Bayesian optimization
 - **Computing:** HPC, GPU computing, GPU resource management, CAE/CFD automation
 - Technical presentation, technical writing, data analysis, and MEMS fabrication experience
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Research Projects

- **Physics-AI:** AI-driven design of automotive vehicles following physics informed machine learning.
 - **Structural Optimization:** Custom multi-objective and surrogate-assisted optimization frameworks for multidisciplinary design of composite aerostructure.
 - **Plant-Inspired Robots:** Designed plant inspired micro air vehicle.
 - **Bio-Inspired Robots:** Designed and developed flapping-wing nano air vehicles using advanced biomimetic technologies.
 - **Renewable Energy Solutions:** Designed catalytic converters and contributed to geothermal power research.
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Awards and Achievements

- Recipient of prestigious **KANAZAWA Memorial** and **JASSO fellowship awards**, Japan.
 - Recognized as a **Finalist in the Young Scientist Conclave** under the "Clean India" theme.
 - Secured top ranks in competitive exams like **GATE** (95.87 percentile) and **IIT** (96.86).
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Volunteer and Leadership Roles

- Delivered lectures on **Robotics and Autonomous systems** in Amity University, Patna.
 - Technical Program Committee Member, **IEEE-ICRAE Conference**.
 - Elected **Hostel Representative** during postgraduate studies, managing various activities.
 - Actively organized technical and cultural events show team-building & coordination skills.
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Publications and Conferences

- Authored **20+ international papers** in peer-reviewed journals and conference proceedings.
 - Presented at prestigious conferences such as **AIAA Aviation**, **SciTech**, and the **IEEE-MEMS**, **IEEE-NEMS**.
 - Key publications on **biomimetics**, and **aerostructural engineering**.
1. **Rashmikant** and Y. Abe, "Integrated aerostructural design of composite aircraft using inertia relief and multi-objective optimization", *Aerospace Science and Technology* vol. 176, 2026 (IF: 5.8).
 2. **Rashmikant** and D. Ishihara, "Flapping wing air vehicles at insect scale: Modeling, design, and fabrication issues", *Robotics and Autonomous Systems*, vol. 202, 2026 (IF: 6.0).
 3. **Rashmikant**, M. Onishi, K. Sugikawa, and D. Ishihara, "Computational Evaluation of Flight Performance of Flapping-Wing Nano Air Vehicles Using Hierarchical Coupling of Nonlinear Dynamic and Fluid-Structure Interaction Analyses", *Engineering Applications of Computational Fluid Mechanics*, vol. 17, 2023 (IF: 8.39).
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Languages

- English: Advanced (TOEIC – 775/990)
 - Japanese: Basic (JLPT N5)
 - Hindi: Native
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Research Interests

Bio-inspired robotics, computational mechanics, Multiphysics and fluid–structure interaction, multidisciplinary optimization, Physics-Informed Machine Learning, autonomous systems, and GPU-accelerated simulation.
